

EXPLORATORY DATA ANALYSIS (EDA)

Project Title:

Retail & E-Commerce Sales Performance Analysis

1. Objective

The objective of Exploratory Data Analysis (EDA) is to analyze the cleaned retail and e-commerce sales dataset to understand business performance, identify trends and patterns, compare different business dimensions, and answer the predefined business questions before developing the Power BI dashboard. This analysis provides a strong foundation for generating business insights and supporting data-driven decision-making.

2. Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Microsoft Power BI
- Power Query

3. Dataset Overview

The Exploratory Data Analysis was performed on the cleaned Orders and Returns datasets after completing data quality assessment and Power Query transformations.

The dataset contains information related to:

- Orders
- Customers
- Products
- Categories
- Sub-Categories
- Brands
- Regions
- States
- Shipping Modes
- Payment Methods
- Sales
- Profit
- Quantity
- Discount
- Returns

The analysis was conducted using Microsoft Excel Pivot Tables and Pivot Charts to answer the predefined business questions and identify meaningful business patterns.

4. Exploratory Data Analysis

Business Question 1

How are sales performing over time?

Analysis

The monthly sales trend was analyzed to understand how sales changed throughout the year. The analysis focused on identifying seasonal trends, monthly performance variations, and overall sales patterns.

Observation

The monthly sales trend shows fluctuations throughout the year, indicating seasonal variations in customer demand and purchasing behavior.

Business Question 2

How does each product category perform in terms of sales and profit?

Analysis

Sales and profit were compared across the three product categories: Electronics, Furniture, and Office Supplies. The analysis evaluated each category's contribution to overall business performance.

Observation

Product categories contribute differently to total sales and profit, indicating variations in revenue generation and profitability.

Business Question 3

Which sub-categories generate the highest sales and profit?

Analysis

Sales and profit were analyzed for each product sub-category to compare their individual business performance.

Observation

Sub-category performance varies across the product portfolio, with some sub-categories contributing significantly more to sales and profit than others.

Business Question 4

Which products generate the highest sales and profit?

Analysis

Individual product performance was analyzed by comparing sales and profit across all products.

Observation

Different products contribute differently to business performance, highlighting variations in customer demand and profitability.

Business Question 5

Which brands contribute the highest sales and profit?

Analysis

Brand-wise sales and profit were analyzed to evaluate the revenue contribution and profitability of each brand.

Observation

Brand performance differs across the business, demonstrating variations in customer preference and brand profitability.

Business Question 6

Which customer segments contribute the most sales and profit?

Analysis

Sales and profit were analyzed across Corporate, Consumer, and Home Office customer segments to understand customer purchasing behavior.

Observation

Each customer segment contributes differently to overall business performance, reflecting differences in purchasing patterns.

Business Question 7

Which regions and states perform the best, and which need improvement?

Analysis

Sales and profit were analyzed across different regions and states to evaluate geographical business performance.

Observation

Business performance varies across different regions and states, indicating opportunities for regional performance improvement.

Business Question 8

Which shipping modes are used most frequently, and how do they impact sales?

Analysis

Shipping modes were analyzed based on sales value and order volume to understand customer shipping preferences.

Observation

Different shipping methods contribute differently to both revenue generation and customer order volume.

Business Question 9

Which payment methods are preferred by customers?

Analysis

Customer payment methods were analyzed based on sales and order count to understand payment preferences.

Observation

Customer payment preferences vary across available payment methods, with digital payment options showing strong adoption.

Business Question 10

How do discounts affect sales and profit?

Analysis

Sales and profit were analyzed across different discount levels to evaluate the relationship between discounts, revenue, and profitability.

Observation

Discount levels influence both sales performance and business profitability, highlighting the importance of an effective pricing strategy.

Business Question 11

Which products are returned most frequently, and what are the return reasons?

Analysis

Returned products and return reasons were analyzed to understand product return patterns and customer return behavior.

Observation

Product returns occur due to multiple reasons, providing opportunities to improve product quality, packaging, and order fulfillment processes.

Business Question 12

What is the overall business performance based on KPIs?

Analysis

Key Performance Indicators (KPIs) such as Total Sales, Total Profit, Total Orders, Total Quantity Sold, Average Discount, Returned Orders, Return Rate, and Profit Margin were analyzed to evaluate overall business performance.

Observation

The KPI analysis provides a comprehensive overview of business performance and supports performance monitoring through the Power BI dashboard.

5. EDA Summary

The Exploratory Data Analysis examined the retail and e-commerce dataset from multiple business perspectives, including sales trends, product categories, sub-categories, products, brands, customer segments, regions, shipping methods, payment methods, discounts, returns, and overall business KPIs. The analysis successfully answered all predefined business questions and established a strong analytical foundation for dashboard development and business decision-making.

6. Conclusion

The Exploratory Data Analysis successfully transformed the cleaned retail dataset into meaningful business information by identifying important trends and performance patterns.

The findings from this phase supported the creation of DAX measures, dashboard visualizations, business insights, and business recommendations. This analysis played a vital role in developing an interactive Power BI dashboard capable of supporting effective data-driven business decisions.